# Farhan Islam

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## Education

**McMaster University** Hamilton, ON

*Bachelor of Engineering, Electrical Engineering (B.Eng.)* *Expected Graduation: Apr 2027*

- **Engineering Award of Excellence**
- Relevant Coursework: Data Structures & Algorithms, Microprocessor Systems, Electronic Devices & Circuits, Signals & Systems, Electromagnetics, Logic Design

## Experience

**Avionics Intern** May 2026 – Present

*Reaction Dynamics* *Longueuil, QC*

- Integrated **flight avionics** for a **hybrid launch vehicle**, verifying power rail stability, sensor interfaces, and communication links against launch readiness specs across each subsystem
- Built and routed **electrical harnesses** across avionics subsystems, catching wiring faults before system-level testing and cutting bench rework
- Designed **PCB layouts** in KiCad and 3D-printed test fixtures for each revision, cutting the time from schematic to validated physical prototype
- Soldered **SMD (0603/0805)** and **through-hole** components, then ran continuity checks and functional validation to confirm boards were flight-ready before integration

**Launch Oversight and Logistics** Sep 2024 – Jan 2025

*McMaster Rocketry Club* *Hamilton, ON*

- Built avionics and telemetry systems for rocket **Nimbus**, driving the team to **1st place in the Payload Challenge** and **3rd place in the Basic Category** at Launch Canada 2025
- Integrated hardware and software across avionics and telemetry subsystems, running rigorous pre-flight testing that resulted in **3 successful rocket** launches without a critical failure
- Developed Rust and C++-based telemetry analysis tools, cutting anomaly detection time by **30%** and accelerating post-launch debugging cycles

**Electrical Team Lead** Sep 2023 – Apr 2024

*Team 9659, FIRST Robotics Competition* *Toronto, ON*

- Led a 5-person team to build a competition-ready robot from scratch, owning the entire **power distribution and motor control architecture**
- Earned the **District Championship Rookie All-Star Award** and **FIRST Ontario Provincial Championship Qualifying Award**
- Redesigned power distribution layout to eliminate mid-match brownouts and voltage drops under peak load

## Projects

**Accessible Microwave Control System** | *Arduino, C++, ESP32, STM32, Bluetooth, WebSocket* |

- Designed an embedded system to help Kimberly, a user with limited motor control, independently operate a microwave using a **dual-microcontroller ESP32 and STM32 architecture** with real-time audio feedback under **30ms** latency via WebSocket and Bluetooth
- Programmed **ESP32** in C++ implementing multi-sensor fusion, digital filtering, and threshold-based validation logic, eliminating false-positive triggers in noisy kitchen environments
- Integrated STM32 for regulated power distribution across subsystems, eliminating voltage instability under load

**DC Motor Speed Controller with PWM** | *Arduino, C++, H-Bridge Driver*

- Built a motor speed controller from scratch to hold $\pm$**2 RPM** accuracy under varying load, using closed-loop PWM with quadrature encoder feedback
- Applied PID tuning to reduce steady-state error by **90%** and improve dynamic response under load
- Implemented H-Bridge control with soft-start ramping to prevent inrush current

## Technical Skills

**Programming**: C/C++, Python, Rust, MATLAB, Assembly
**Embedded Systems**: ESP32, STM32, Arduino, I2C/SPI/UART, Real-Time Processing
**Power & Control**: PWM Control, PID Tuning, Power Distribution
**Communication**: WebSocket, Bluetooth, Serial Communication
**Tools**: KiCad, LTSpice, AutoCAD, Circuit Prototyping
**Version Control**: Git, GitHub